
BDG: Batch-Dispersion Guidance for Property-Band Generation

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Abstract

Property-targeted generation must satisfy a tolerance band while preserving quality. Standard gradient guidance scales target centering and distribution contraction together. We introduce Batch-Dispersion Guidance (BDG), which adjusts the deviation coefficient using batch endpoint variance relative to a requested variance, retaining the centering coefficient. BDG requires the same neural calls as plug-in guidance. We study QM9 molecules in Euclidean coordinates and DeepFlyBrain DNA on a probability simplex. Our trained flow-matching model achieves 39.7 percent molecular stability versus 28.8 percent for our variance-preserving diffusion model, supporting the choice of flow matching. Plug-in guidance improves dipole and gap coverage on both trained models. Across two flow backbones and three molecular properties, contracting BDG raises mean decoded coverage by 0.45–1.52 percentage points over plug-in and reduces mean absolute target error in all six comparisons. The coverage gains remain below the protocol threshold and have greater seed variability on our model. A 612-cell ablation demonstrates monotone spread adjustment across four setpoints, from 0.67–0.77 to 1.15–1.25 times unguided spread at fixed strength and gain. On DNA, CpG gains reach 4.82 and 16.83 points at strengths one and four. A follow-up GC window starting at $t = 0.3$ separates the baselines and yields a 7.08-point BDG gain at strength four. Hamming diversity changes little; fidelity costs depend on property and window. Four recent guidance adaptations contextualize the concentration–spread trade-off and its property-dependent quality costs.

1 Introduction

Diffusion and flow matching generate structured samples by reversing corruption or learning transport velocities [1, 2]. Comparing them on one task grounds the choice of generator in measured quality and cost [3].

Gradient guidance moves properties toward a target, but its strength also contracts their spread [4, 5]. Batch interactions and moment constraints motivate a separate dispersion setting [6, 7]. Can this setting improve band coverage at an acceptable quality cost?

We compare linear FM and VP diffusion on QM9, then test BDG against plug-in and recent adaptations before transfer to DeepFlyBrain DNA. Our hypothesis is that a scalar property-dispersion rule adjusts coverage across different state spaces. Our contributions are summarized as follows.

- We select FM from a trained-model quality comparison.
- We derive BDG’s zero-gain limit and expansion condition.
- We isolate gain, setpoint and strength in 612 ablation cells.
- We quantify coverage, quality and cost across two modalities.

2 Related Work

Continuous generation. FM fits transport velocities; VP diffusion predicts corruption noise. Stochastic interpolants connect these formulations [2, 3].

Dispersion and guidance. Particle Guidance introduces batch interactions [6]; Moment Guided Diffusion imposes moment constraints [7]; variance tilting favors feature spread [8]. BDG adds a batch variance residual to endpoint guidance, distinct from per-sample, uncertainty and Monte Carlo corrections. We claim this composition without distributional guarantees.

Molecules. Equivariant diffusion and EquiFM model molecular geometry [9, 10]. We compare DPS-style plug-in, TMPD-inspired guidance, LGD-MC and TFG adaptations on common frozen backbones [4, 11, 5, 12].

Sequences. Dirichlet FM and Fisher Flow model simplex-valued sequences; MOG-DFM guides discrete biological sequences [13–15]. Our measured sequence comparators are ports of the guidance rules above, not reproductions of these three generators.

3 Methods

3.1 Modalities, Data, and Preprocessing

QM9 [16] uses centered coordinates and one-hot H/C/N/O/F. Generator/guide f_A and evaluator f_B use disjoint 51,527-molecule halves; validation/test have 17,748/13,083. Targets are dipole μ , polarizability α , and HOMO–LUMO gap. DeepFlyBrain has 83,722/10,505/10,434 train/validation/test 500-bp sequences on $(\Delta^3)^{500}$. Generators are property-unconditional; molecules use atom-count mask M (Appendix A.1).

3.2 Continuous Flow Matching Baseline

With Gaussian source x_0 , data x_1 and uniform $t \in [0, 1]$, we train

$$x_t = (1 - t)x_0 + tx_1, \quad \mathcal{L}_{\text{FM}} = \mathbb{E} \|v_\theta(x_t, t; M) - (x_1 - x_0)\|_M^2. \quad (1)$$

The 3.75M-parameter EGNN (8 layers, width 256) uses Adam, batch 256, rate 2×10^{-4} , cosine decay and EMA 0.9999. Epoch-1500 EMA sampling uses 100 Euler steps (Table S1).

3.3 Diffusion Baseline

Our trained VP model uses the same representation and EGNN configuration. For corruption time u , $a(u) = \exp[-(0.1u + 9.95u^2)/2]$ and $\sigma^2(u) = 1 - a^2(u)$:

$$x_u = a(u)x_1 + \sigma(u)\epsilon, \quad \mathcal{L}_{\text{VP}} = \mathbb{E} \|\epsilon_\phi(x_u, u; M) - \epsilon\|_M^2. \quad (2)$$

Sampling integrates $dx/du = -\frac{1}{2}\beta(u)(x - \epsilon_\phi/\sigma)$ backward, with $\beta(u) = 0.1 + 19.9u$, 100 steps and terminal denoising. We evaluate the selected epoch-1475 EMA checkpoint.

3.4 Guided Generation

The frozen guide evaluates $F_i = f_A(m_i)$ at $m_i = x_t^{(i)} + (1 - t)v_\theta(x_t^{(i)}, t; M)$. Let s be the calibration-pool property standard deviation. For target y , plug-in guidance adds

$$G_i = \frac{y - F_i}{s^2} \nabla_{x_t^{(i)}} F_i, \quad C_i = \text{clip}_{\|v_i\|} \left[w \frac{1 - t}{\max(t, \varepsilon)} G_i \right]. \quad (3)$$

On QM9 we apply C_i for $t \geq 0.5$, then mask and recenter molecular coordinates. The derivative includes the generator’s endpoint Jacobian. All model weights stay frozen.

3.5 Proposed Innovation: Batch-Dispersion Guidance

For $B > 1$, define $\bar{F} = B^{-1} \sum_i F_i$, $V_b = (B - 1)^{-1} \sum_i (F_i - \bar{F})^2$, gain $\eta \geq 0$ and setpoint $\tau = \tau_{ms} > 0$. BDG replaces $y - F_i$ by

$$\begin{aligned} n_i &= (y - F_i) - \eta e(F_i - \bar{F}) = (y - \bar{F}) - w_{\text{eff}}(F_i - \bar{F}), \\ e &= (V_b - \tau^2)/\tau^2, \quad w_{\text{eff}} = 1 + \eta e. \end{aligned} \quad (4)$$

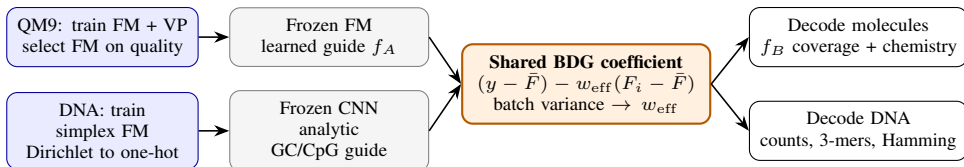


Figure 1: Training (blue), frozen components (gray), and shared BDG rule (orange). The state, guide and decoder change across modalities; the scalar dispersion rule is unchanged.

BDG retains the centering coefficient and detaches n_i/s^2 before pullback. At $\eta = 0$, **BDG equals plug-in**; $\tau = 0$ is undefined. Since $V_b \geq 0$, $w_{\text{eff}} \geq 1 - \eta$. Negative e weakens contraction; expansion requires $w_{\text{eff}} < 0$, hence $\eta > 1$. Terminal spread is not guaranteed (Algorithm 1).

3.6 Transfer to Modality 2

A ten-layer, width-128 CNN learns a Dirichlet(1, 1, 1, 1)-to-one-hot linear path. Training ran 1,500 epochs; validation selected epoch 1,450. BDG uses analytic soft GC/CpG properties. Euler steps clamp and renormalize; final decoding uses arg max (Appendix A.7).

3.7 Model Overview

Figure 1 separates training, frozen guidance and decoding across the two state spaces.

4 Results

4.1 Evaluation Protocol

4.1.1 Metrics

Targets are training medians (q50). Molecular IB counts decoded samples within $\delta = 2 \text{MAE}_{\text{cal}}(f_B)$ of y , accompanied by stability, validity and distinct-valid yield. DNA counts exact GC/CpG with half-widths 0.008833/0.008818, plus 3-mer JSD and Hamming diversity. Figure S1 shows training distributions, q50 and both band rules. Coverage comparisons stay within matched tasks.

4.1.2 Fair Comparison and Computational Budget

Settings use three seeds, 2,000 samples each, batches of 500 and 100 Euler steps. Headline BDG uses $\eta = 4$, $w = 1$; ablations and DNA also use $w = 4$. DNA retains $t \geq 0.5$ and adds a GC-only $t \geq 0.3$ follow-up at $w = 4$; windows are never pooled (Appendix A.4).

4.1.3 Statistical Reporting

Tables show mean $\pm$ seed sd. Molecular comparisons use unpaired binomial $|z| \geq 2.99$ (18 contrasts, family level 0.05), ignoring batch coupling and rerun noise (Table S8). Original DNA uses an exploratory $z = 3.16$ threshold for 32 contrasts; the new window is descriptive.

4.2 Modality 1: Flow Matching versus Diffusion

FM’s higher stability (39.7% versus 28.8%) and validity (75.6% versus 66.2%) than our VP diffusion justify carrying it forward (Table 1). Pretrained EquiFM (flow matching) also exceeds EDMsecond (diffusion) on these metrics. Our models share data, architecture and training settings; selection differs (Appendix A.3). External checkpoints provide supporting context.

4.3 Modality 1: Guided versus Unguided Generation

Plug-in raises decoded dipole/gap coverage by 2.75/2.35 points on FM and 2.13/2.60 on VP, all above threshold; both α changes remain unresolved. Stability falls by 4.08/6.15 points on FM and 2.68/5.65 on VP (Tables 3, S7). Guidance therefore improves targeting on both families with a chemistry cost.

Table 1: Unguided QM9, $3 \times 2,000$ samples: percentages, mean $\pm$ seed sd. Lower rows use pretrained external models. Timing includes scoring: RTX A6000 except *B200 MIG; runtimes are not hardware matched.

Generator	Mol. stable $\uparrow$	Valid $\uparrow$	Unique $\uparrow$	NFE	s/sample
FM, ours	39.7 ± 0.6	75.6 ± 1.1	99.8	100	0.051
VP diffusion, ours	28.8 ± 1.3	66.2 ± 0.7	99.9	101	0.048*
EquiFM (flow)	86.2 ± 0.3	93.8 ± 0.6	99.7	100	0.086
EDMsecond (diffusion)	67.2 ± 1.3	86.0 ± 1.3	99.8	101	0.087

4.4 Modality 1: Innovation Ablations

Table 2 removes BDG’s residual ($\eta = 0$), changes its setpoint and increases plug-in strength. Contracting BDG improves dipole target concentration; increasing τ_m widens spread past unguided, which neither tested plug-in strength reaches (Figure 2a). The 612-cell grid and descriptive quality-cost comparisons appear in Appendix A.6 and Table S9.

Table 2: Dipole ablation on our FM: q50 $y = 2.4932$ D, band ± 0.17541 D. IB is percent mean $\pm$ seed sd; stability is percent mean. Decoded MAE, mean and spread σ are in D. Zero gain is plug-in; its $w = 4$ row tests stronger guidance.

w	η	τ_m	IB $\uparrow$	MAE $\downarrow$	Mean	σ	Stable $\uparrow$
1	0	–	10.6 ± 0.4	0.969	2.615	1.209	35.6
4	0	–	13.4 ± 0.4	0.812	2.548	1.024	30.3
1	4	0.5	11.2 ± 1.1	0.903	2.609	1.113	35.1
1	4	1	8.9 ± 0.2	1.133	2.714	1.434	40.5
1	8	0.5	11.9 ± 0.3	0.861	2.603	1.064	34.0
1	8	1.5	7.6 ± 0.2	1.377	2.858	1.922	37.6

4.5 Modality 1: Comparison with Recent Methods

At $\tau_m = 0.5$, BDG improves mean coverage and MAE in all six flow/property comparisons, with higher seed variability on our FM and unresolved coverage gains. At $\tau_m = 1$, spread increases and occupancy falls. BDG matches plug-in’s neural calls and uses one fifth of LGD-MC/TFG’s guide calls (Table S3). TFG achieves higher coverage with greater stability loss (Tables 3, S4–S6).

Table 3: Our FM, $w = 1$: decoded IB, percent mean $\pm$ seed sd; stability means in $\mu/\alpha/\text{gap}$ order. DV is distinct-valid yield (%), and time is s/sample, both for μ . Plug-in is $\eta = 0$; BDG uses $\eta = 4$. Rows are matched-backbone adaptations.

Method	IB $\mu \uparrow$	IB $\alpha \uparrow$	IB gap $\uparrow$	Stable $\uparrow$	DV $\uparrow$	Time
Unguided	7.8 ± 0.1	5.0 ± 0.3	10.3 ± 1.0	39.7 / 39.7 / 39.7	75.5	0.051
Plug-in ($\eta = 0$)	10.6 ± 0.4	6.1 ± 0.3	12.6 ± 0.7	35.6 / 37.2 / 33.6	72.4	0.119
TMPD-inspired	9.0 ± 0.6	6.1 ± 0.4	11.6 ± 0.9	38.4 / 38.9 / 37.3	74.7	0.255
LGD-MC	9.6 ± 0.8	5.2 ± 0.4	10.8 ± 0.7	39.9 / 39.2 / 39.5	75.4	0.229
TFG	28.8 ± 1.6	10.5 ± 0.3	27.5 ± 0.3	24.5 / 26.8 / 14.4	62.7	0.122
BDG , $\tau_m = 0.5$	11.2 ± 1.1	7.0 ± 0.5	14.2 ± 0.8	35.1 / 36.4 / 30.0	71.2	0.119
BDG , $\tau_m = 1$	8.9 ± 0.2	5.6 ± 0.5	10.6 ± 0.2	40.5 / 37.9 / 37.6	76.3	0.119

4.6 Modality 2: Transfer of the Innovation

The new GC follow-up starts guidance at $t = 0.3$: at $w = 4$, contracting BDG raises coverage from 30.25% to 37.33% (+7.08 points); the four baselines now differ. Its JSD is below unguided but above plug-in. CpG remains at $t \geq 0.5$, with a +16.83-point gain and higher JSD (Table 4). Hamming diversity changes little. Original $t \geq 0.5$ GC/CpG results remain in Table S17; only that late window gives identical plug-in/TMPD/LGD coverage on both properties.

Table 4: DNA, $w = 4$: new GC window $t \geq 0.3$; CpG remains $t \geq 0.5$. IB is percent mean $\pm$ seed sd; JSD ($\times 10^{-4}$), Hamming and s/sequence are GC/CpG. Dashes denote an unrun GC arm. Original matched-window results: Table S17.

Method	GC IB $\uparrow$	CpG IB $\uparrow$	JSD $\downarrow$	Hamming	Time
Unguided	14.0 ± 1.0	50.2 ± 1.6	4.53 / 4.53	0.7456 / 0.7456	0.021 / 0.049
DPS-style plug-in	30.3 ± 2.0	51.6 ± 1.4	3.64 / 4.63	0.7456 / 0.7456	0.124 / 0.096
TMPD-inspired	29.2 ± 2.1	51.6 ± 1.4	3.66 / 4.63	0.7455 / 0.7456	0.145 / 0.124
LGD-MC	30.0 ± 1.3	51.6 ± 1.4	3.61 / 4.63	0.7455 / 0.7456	0.131 / 0.102
TFG-MC adaptation	29.8 ± 1.8	51.1 ± 1.7	3.57 / 4.58	0.7455 / 0.7456	0.109 / 0.084
BDG , $\tau_m = 0.5$	37.3 ± 6.5	68.5 ± 1.2	4.24 / 7.73	0.7455 / 0.7457	0.122 / 0.076
BDG , $\tau_m = 1$	—	50.3 ± 1.5	— / 4.52	— / 0.7456	— / 0.076

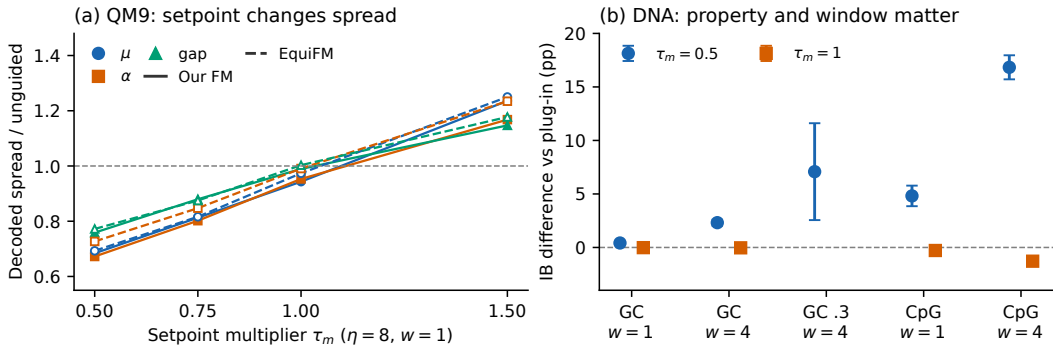


Figure 2: (a) Decoded property spread relative to each backbone’s unguided reference; points pool three seeds. Solid: our FM; dashed: EquiFM. (b) DNA coverage gain over plug-in, $\eta = 4$; bars are sd of three paired seed differences. QM9 uses $t \geq 0.5$. DNA uses $t \geq 0.5$ except GC .3 ($t \geq 0.3$); no $\tau_m = 1$ cell was run there.

4.7 Cross-Modality Analysis and Failure Modes

BDG adjusts spread in both modalities. GC baselines separate with earlier guidance, supporting a timing explanation for their late-window overlap. Different bands, windows and clipping prevent a curvature-only explanation for GC/CpG gains (Appendix A.7).

5 Discussion

FM provides the stronger QM9 starting point; guidance improves targeting on both families. BDG adds adjustable spread and higher mean target concentration at plug-in’s neural cost. TFG achieves more coverage with greater chemistry loss. GC and CpG gains extend BDG to a second state space.

The zero-gain and setpoint ablations connect BDG’s added coefficient to decoded spread and occupancy. Retaining the centering term lets dispersion change without rescaling that coefficient; this does not imply independent mean dynamics or faster convergence.

Batch-aware intervals and joint in-band-and-stable yield would strengthen evaluation; fidelity-matched DNA comparisons would quantify the cost of CpG gains. Matched-force comparisons and open-loop replay would test whether feedback itself is necessary. Next, control molecular atom counts more closely and define evaluation goals around the local property distribution near each target range, to address size and target-density confounding.

References

- [1] T. Karras, M. Aittala, T. Aila, and S. Laine, “Elucidating the design space of diffusion-based generative models,” *Advances in neural information processing systems*, vol. 35, pp. 26565–26577, 2022.
- [2] Y. Lipman, R. T. Chen, H. Ben-Hamu, M. Nickel, and M. Le, “Flow matching for generative modeling,” *arXiv preprint arXiv:2210.02747*, 2022.
- [3] M. Albergo, N. M. Boffi, and E. Vanden-Eijnden, “Stochastic interpolants: A unifying framework for flows and diffusions,” *Journal of Machine Learning Research*, vol. 26, no. 209, pp. 1–80, 2025.
- [4] H. Chung, J. Kim, M. T. McCann, M. L. Klasky, and J. C. Ye, “Diffusion posterior sampling for general noisy inverse problems,” in *International Conference on Learning Representations (ICLR)*, 2023. arXiv:2209.14687.
- [5] J. Song, Q. Zhang, H. Yin, M. Mardani, M.-Y. Liu, J. Kautz, Y. Chen, and A. Vahdat, “Loss-guided diffusion models for plug-and-play controllable generation,” *Proceedings of the 40th International Conference on Machine Learning (PMLR v202)*, 2023. <https://proceedings.mlr.press/v202/song23k.html>.
- [6] G. Corso, Y. Xu, V. De Bortoli, R. Barzilay, and T. Jaakkola, “Particle guidance: Non-I.I.D. diverse sampling with diffusion models,” in *International Conference on Learning Representations (ICLR)*, 2024. arXiv:2310.13102.
- [7] E. Lempereur, N. Cuvelle-Magar, F. Coeurdoux, S. Mallat, and E. Vanden-Eijnden, “MGD: Moment guided diffusion for maximum entropy generation,” *arXiv preprint arXiv:2602.17211*, 2026.
- [8] I. Azangulov, L. Zhang, and K. Ashouritaklimi, “Variance-tilted diffusion models for diverse sampling,” *arXiv preprint arXiv:2606.22239*, 2026.
- [9] E. Hoogeboom, V. G. Satorras, C. Vignac, and M. Welling, “Equivariant diffusion for molecule generation in 3d,” in *International conference on machine learning*, pp. 8867–8887, PMLR, 2022.
- [10] Y. Song, J. Gong, M. Xu, Z. Cao, Y. Lan, S. Ermon, H. Zhou, and W.-Y. Ma, “Equivariant flow matching with hybrid probability transport,” *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.
- [11] B. Boys, M. Girolami, J. Pidstrigach, S. Reich, A. Mosca, and O. D. Akyildiz, “Tweedie moment projected diffusions for inverse problems,” *Transactions on Machine Learning Research (TMLR)*, 2024. arXiv:2310.06721.
- [12] H. Ye, H. Lin, J. Han, M. Xu, S. Liu, Y. Liang, J. Ma, J. Zou, and S. Ermon, “TFG: Unified training-free guidance for diffusion models,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2024. arXiv:2409.15761.
- [13] H. Stark, B. Jing, C. Wang, G. Corso, B. Berger, R. Barzilay, and T. Jaakkola, “Dirichlet flow matching with applications to DNA sequence design,” in *Proceedings of the 41st International Conference on Machine Learning*, vol. 235, pp. 46495–46513, PMLR, 2024.
- [14] O. Davis, S. Kessler, M. Petrache, I. I. Ceylan, M. Bronstein, and A. J. Bose, “Fisher flow matching for generative modeling over discrete data,” *Advances in Neural Information Processing Systems*, vol. 37, pp. 139054–139084, 2024.
- [15] T. Chen, Y. Zhang, S. Tang, and P. Chatterjee, “Multi-objective-guided discrete flow matching for controllable biological sequence design,” *arXiv preprint arXiv:2505.07086*, 2025.
- [16] R. Ramakrishnan, P. O. Dral, M. Rupp, and O. A. von Lilienfeld, “Quantum chemistry structures and properties of 134 kilo molecules,” *Scientific Data*, vol. 1, p. 140022, 2014.
- [17] V. G. Satorras, E. Hoogeboom, and M. Welling, “E(n) equivariant graph neural networks,” in *Proceedings of the 38th International Conference on Machine Learning*, vol. 139 of *Proceedings of Machine Learning Research*, pp. 9323–9332, 2021.

A Reproducibility and Extended Results

A.1 Data Sources and Splits

QM9 uses the DeepChem/MoleculeNet packaging. The project retains the 3,054 entries excluded as uncharacterized in common EDM preprocessing, so published EDM dataset definitions are not identical. A seeded permutation creates the four splits stated in Section 3.1. Coordinates are centered, padded atoms are masked, and coordinates and one-hot features have no additional scaling. The E(3)-equivariant generator handles rotation and translation; invariant property predictors use training-derived normalization. Sample i uses validation mask i for $i = 0, \dots, 1999$, identically across every arm, seed and backend. This fixed list has minimum 8, median 18, mean 17.87 and maximum 29 atoms. No histogram draw, size filtering or target-dependent size assignment occurs; all sizes receive the same q50 target. Seeds change initial noise only. No property label enters either generator.

DeepFlyBrain is the 500-bp enhancer data used by Dirichlet FM [13]. Four ambiguous training examples were excluded from the supplied 83,726. The retained splits are 83,722/10,505/10,434; channels are A/C/G/T. Class labels are unused by our unconditional generator. These random-example splits do not test sequence-cluster generalization. The older 200-bp experiment is excluded.

Exact targets and calibration values are preserved in each source cell. Molecular band half-widths for our predictor pair are 0.17541 D for μ , 0.50618 Bohr³ for α , and 0.00748 Ha for gap. External EquiFM uses a different predictor pair and therefore different acceptance widths. Its coverage must not be ranked directly against our FM’s coverage.

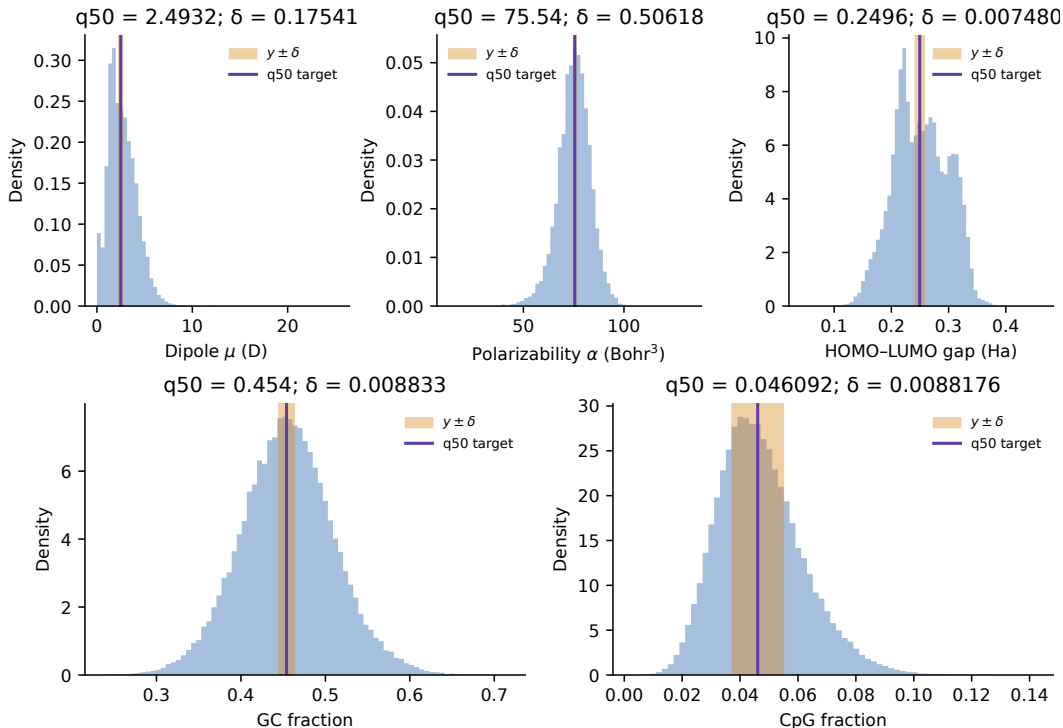


Figure S1: Measured training-property distributions: QM9 train_a (51,527 molecules, top) and DeepFlyBrain training (83,722 sequences, bottom). Purple marks q50 y ; orange marks $[y - \delta, y + \delta]$. Histogram bins cover the full observed support. These are real-data reference distributions, not generated samples. QM9 bands shown use our predictor pair; EquiFM uses its separate calibrated bands.

Evaluation band versus controller setpoint. For QM9,

$$\delta = 2 \text{MAE}_{\text{cal}}(f_B) = \frac{2}{|\mathcal{C}_B|} \sum_{j \in \mathcal{C}_B} |f_B(x_j) - y_j|.$$

The internal evaluator is scored on calibration examples from train_a, which it did not train on. For DNA, $\delta = \max(0.16s, 4.4q)$, where s is the training-property standard deviation and $q = 1/500$ (GC) or $1/499$ (CpG). The lattice floor binds for CpG. Full bandwidth is 2δ ; δ is fixed across guidance arms. BDG varies the requested spread $\tau = \tau_m s$, not the acceptance band.

A.2 EGNN, Predictor Roles, and Molecular Decoding

EGNN stands for E(n)-Equivariant Graph Neural Network [17]. We implement its layers in plain PyTorch and train every internal generator and property network from scratch. EquiFM and EDM checkpoints are separate borrowed backends. Every distinct pair of real atoms is connected; padding and self-pairs are masked. E(n) includes rotations, reflections and translations. Distances are invariant, difference vectors transform with the molecule, and summing neighbours respects atom permutations. The scalar dipole magnitude is invariant; a dipole vector would rotate.

For atom features h_i , positions x_i and $d_{ij} = \sqrt{\|x_i - x_j\|^2 + 10^{-8}}$, a layer computes

$$\begin{aligned} m_{ij} &= \phi_e(h_i, h_j, \|x_i - x_j\|^2), \\ h_i &\leftarrow h_i + \phi_h(h_i, \sum_j m_{ij}), \\ x_i &\leftarrow x_i + \sum_j (x_i - x_j) \phi_x(m_{ij}) / (d_{ij} + 1). \end{aligned}$$

Each ϕ is an MLP. Messages describe pairs, their sum updates features, and learned scalar weights move atoms along pair directions. One dense layer reaches all atoms; stacking layers learns richer interactions.

EGNNVelocity is the generator. Its input is noisy coordinates, continuous type channels, time and mask. Coordinate output is final minus initial position, recentered; a feature MLP returns type-channel output. FM reads these as velocities and VP as noise estimates. EGNNScalar keeps coordinates fixed, averages final features over real atoms, applies a pooling MLP, and uses a linear head for one property. Both f_A and f_B use this class with separate weights: f_A steers predicted endpoints through the generator Jacobian; f_B evaluates decoded samples and is absent from the guidance update. The training halves and calibration roles are disjoint as specified above.

For one molecule the FM error $z = v_\theta - v^*$ is 29×8 : z^c is 29×3 coordinate-velocity error and z^f is 29×5 type-velocity error. Types remain continuous during noising and sampling; at the data endpoint they are one-hot H/C/N/O/F. Thus $v = (v^c, v^f)$ changes positions and type numbers. The loss masks padding and normalizes each part separately, as below.

From cloud to molecule. (1) The generator outputs real atom positions and five continuous type channels, with no bonds. (2) Argmax assigns the element: $[0.02, 0.97, 0.01, 0, 0.03] \mapsto C$. (3) Pair distances in Å are multiplied by 100 and compared with EDM’s bond-length tables in pm, with margins 10/5/3 for single/double/triple bonds. For C–C, references are 154/134/120 pm: triple for $d < 123$, double for $123 \leq d < 139$, single for $139 \leq d < 164$, and no bond otherwise. (4) Bond-order sums are checked against H:1, C:4, N:3, O:2, F:1; molecular stability requires all real atoms to pass. (5) RDKit builds atoms and bonds, sanitizes and canonicalizes SMILES. (6) Independently, f_B scores final positions and argmax one-hot types for in-band membership; it does not consume the inferred bonds.

A.3 Training and Checkpoint Records

Table S1 records the training settings and evaluated checkpoints. For coordinates c , features f and broadcast mask M , the training norm is

$$\|z\|_M^2 = \frac{\|M \odot z^c\|_2^2}{3 \sum M} + \frac{\|M \odot z^f\|_2^2}{5 \sum M}.$$

Gaussian coordinate noise is centered; feature noise is independent Gaussian. Both molecular checkpoints specify the same split, architecture, optimizer, batch size, 1,500-epoch budget, EMA and training seed. The evaluated FM export is epoch-1500 EMA; VP is the selected epoch-1475 EMA, with validation loss 0.20784. The trainer selects validation atom stability with validation loss as a

tie-breaker. The VP export records the selected epoch and configured budget, not proof of the final completed epoch. Consequently, Table 1 compares these trained checkpoints under shared settings with a disclosed selection difference; it does not establish family-wide superiority. VP training samples $u \in [10^{-3}, 1]$; its probability-flow ODE follows Equation (2).

Table S1: Training records and evaluated checkpoints. VP’s selected epoch is 1475 within a configured 1500-epoch budget. Molecular predictors use one network per property and training half. Unrecorded training time is not inferred from sampling time.

Setting	Molecular FM / VP	f_A / f_B	Sequence FM
Architecture	EGNN, 256, 8 layers	EGNN, 128, 4	CNN, 128, 10
Parameters	3.75M / 3.75M	429,825 each	approximately 1.02M
Training data	<code>train_a</code>	<code>train_a/train_b</code>	83,722 sequences
Objective	velocity / noise MSE	property MSE	velocity MSE
Epoch budget	1500 / 1500	120	1500 completed
Evaluated weights	EMA: 1500 / 1475	validation checkpoint	epoch 1450
Batch size	256	128	256
Optimizer	Adam	Adam	Adam
Initial learning rate	2×10^{-4}	3×10^{-4}	2×10^{-3}
Schedule	cosine, floor 5%	cosine	cosine to zero
EMA	0.9999	none	none
Training seed	20260918 / 20260918	20260918	20260921
Training time	not fully recorded	not fully recorded	539.7 min

Sequence training performed 492,000 updates across 1,500 epochs. Validation selected update 475,600 (epoch 1450), loss 0.0633644, with 121.75M sample-views at the selected checkpoint. The total run lasted 539.7 minutes on RTX A6000, including validation. We distinguish completed training length from the selected checkpoint’s age. The checkpoint uses Adam without EMA; its best-validation weights are sampled.

Table S2: Checkpoint fingerprints from the result records and local files. Prefixes identify artifacts; full hashes are retained in the source records.

Component	Identifier	Evidence
Our molecular FM	MD5 e19ccc06	v3 cells, epoch 1500
Borrowed EDMsecond	MD5 6abbd010	v3 cells, released checkpoint
Sequence FM	MD5 7f59b60b	local checkpoint, best validation
Our VP diffusion	MD5 8a3390a6	v3 cells, epoch 1475

Table S2 identifies the evaluated artifacts. FM, EquiFM and EDMsecond cells record PyTorch 2.11.0+cu130, CUDA 13.0 and RTX A6000. VP cells record PyTorch 2.11.0+cu128, CUDA 12.8 and NVIDIA B200 MIG 2g.45gb. VP’s local checkpoint MD5 matches every result cell. Hardware differs, so elapsed seconds cannot establish an FM–VP speed advantage. The sequence training report identifies RTX A6000, but individual sequence evaluation cells omit the device; those timings are not hardware-matched cross-modality benchmarks.

A.4 Evaluation, Uncertainty, and Cost

Molecular seeds are 20261001, 20261002 and 20261003. Sequence seeds are 20260921, 20260922 and 20260923. Each 2,000-sample cell contains four independent controller batches of size 500. Enlarging the total sample count adds batches, not observations to a single variance estimate. At fixed weights, seed variation measures sampling variation, not training uncertainty.

Molecular coverage counts decoded samples satisfying $|f_B(\hat{x}) - y| \leq \delta$. Calibration uses 3,000 molecules drawn from the two training halves; each internal predictor is scored on the half it did not train on. The wrapper sets s to the property standard deviation of this calibration pool, which differs from the raw predictor checkpoint’s normalization scale. This s sets the guidance energy and BDG setpoint; evaluator error sets δ . Coverage on continuous features remains in source records but is not substituted for decoded IB.

Atom stability checks bond/valence rules; molecular stability requires every atom to pass. RDKit validity checks reconstruction and sanitization. Uniqueness divides distinct valid SMILES by valid samples; distinct-valid yield divides by all attempts. None is an energy or experimental viability measurement. IB includes invalid molecules, so usable yield requires the joint event of decoded in-band membership and stability. Marginal IB and stability cannot be multiplied to recover that event. The local aggregate results do not include the sample-level sidecars needed for this joint analysis.

The reported unpaired statistic is

$$z = \frac{p_a - p_b}{\sqrt{p_a(1 - p_a)/6000 + p_b(1 - p_b)/6000}}.$$

The registered molecular bar is 2.99. Table S8 reports the corresponding threshold-sized difference separately for every contrast; it is neither a power calculation nor evidence of equivalence. Within-batch dependence can invalidate an independent-Bernoulli variance, and common-noise pairing changes covariance. We retain this protocol statistic with explicit limitations and show seed standard deviations. A batch-aware paired bootstrap would need the sample-level records. At $\tau_m = 0.5$, all six molecular coverage differences favor BDG in their means but remain below this bar. At $\tau_m = 1$, three differences cross the bar against BDG and three remain unresolved. These outcomes distinguish an occupancy-setting effect from a universal coverage improvement. DNA uses the rule only as an exploratory comparison, and the stated positive effects survive the 32-contrast threshold of 3.16. For selections from the full molecular grid, the protocol’s adjusted bar is 3.49; the paper makes descriptive grid claims without converting them into headline tests.

EquiFM same-seed reruns differ by up to 0.007 in continuous coverage. Its $\eta = 0$ control agrees with plug-in only to the rerun tolerance, not bitwise. Our FM reproduces exactly. EDMsecond has same-seed variation up to 0.0015 in molecular stability. Neither binomial errors nor three seed values fully characterize this extra implementation variability.

The molecular run contains 162 headline cells, including 18 from our VP diffusion model, and 612 ablation cells. Summed recorded wall-clock is 16.77 hours for headline and 65.15 for ablations, 81.92 total. VP contributes 0.813 hours on different hardware. The original DNA files contain 84 headline, 111 ablation/window and four single-seed strength cells (199 files, 5.26 recorded hours). The new window follow-up adds 22 files: 18 at $t_{\min} = 0.3$ (six arms, three seeds) and four single-seed $t_{\min} = 0.2$ probes. The latter are exploratory, not three-seed comparisons. The new files add 1.60 recorded hours, for 6.87 DNA hours in total. Twelve original configurations repeat across trees; zero-gain controls are reported separately. The repeated configurations are checked for agreement and never pooled as extra seeds. Concurrent cells and evaluation overhead prevent interpreting these totals as elapsed time or accelerator utilization.

NFE counts solver field evaluations, distinct from generator forwards and derivative calls. For each molecular flow batch, plug-in and BDG record 200 generator forwards, 50 generator VJPs, and 50 guide forward/backward pairs; unguided records 100 generator forwards. VP unguided sampling adds terminal denoising for 101 evaluations. Sequence plug-in/BDG batches record 150 generator forwards and 50 VJPs at $t \geq 0.5$, or 170 forwards and 70 VJPs at $t \geq 0.3$; both use 100 solver steps. BDG adds scalar reductions to plug-in, not another neural call. Equal calls do not imply equal measured runtime. Clip fractions divide clipped sample-steps by samples times guided steps *per batch*; using the batch-summed step count understates them fourfold.

Table S3 separates the recorded operation types. BDG and plug-in share these counts, while LGD-MC and TFG use five times as many guide forward/backward calls. These counts describe the sampled configurations; neither derivative calls nor Monte Carlo calls are interchangeable units of runtime.

A.5 Full Molecular Comparisons

Tables S4, S5, S6 and S7 report every headline arm and property. All numbers are generated by this project; no published score is mixed into these rows. EquiFM uses the released TFG guide/evaluator pair; our FM, our VP diffusion and borrowed EDMsecond use our disjoint pair. External training overlap, architectures, scales and selection differ, so all guidance contrasts stay within a backend. Equal nominal strength exposes each adaptation’s behavior under shared settings; it is not a ranking of separately optimized methods.

Table S3: Recorded sampling operations per 2,000-molecule cell, our FM, dipole, $w = 1$. Forward calls, vector–Jacobian products (VJP) and Jacobian–vector products (JVP) are separate operations, not equal-cost units. Evaluator calls are outside these counts; Table 3 reports wall-clock including scoring.

Method	Gen. forward	Gen. VJP	Gen. JVP	Guide forward	Guide backward
Unguided	400	0	0	0	0
DPS-style plug-in	800	200	0	200	200
TMPD-inspired	800	200	200	200	200
LGD-MC	800	200	200	1000	1000
TFG	600	200	0	1000	1000
BDG , $\tau_m = 0.5$	800	200	0	200	200
BDG , $\tau_m = 1$	800	200	0	200	200

DPS-style plug-in adapts posterior guidance to a Gaussian property energy and a flow endpoint. TMPD-inspired guidance uses a linearized uncertainty correction; LGD-MC marginalizes a local property loss; TFG combines training-free corrections. The implementations retain the local endpoint, clipping and timing choices in the cell records, and are adaptations of the cited methods, not reproductions of every original experimental setting. Molecular TFG and sequence TFG-MC are different implementations; the latter omits full TFG’s iterative schedule.

Table S4: Complete headline evaluation on our FM. IB is decoded; IB and stability show percent mean $\pm$ seed sd. Validity and distinct-valid yield (DV) are percentages; time is recorded cell wall-clock divided by attempts. $w = 1$, $t \geq 0.5$, three seeds, $n = 2,000$ each.

Property	Method	IB $\uparrow$	Stable $\uparrow$	Valid $\uparrow$	DV $\uparrow$	s/sample
μ	Unguided	7.8 ± 0.1	39.7 ± 0.6	75.6	75.5	0.051
μ	DPS-style plug-in	10.6 ± 0.4	35.6 ± 0.3	72.5	72.4	0.119
μ	TMPD-inspired	9.0 ± 0.6	38.4 ± 0.6	74.8	74.7	0.255
μ	LGD-MC	9.6 ± 0.8	39.9 ± 0.3	75.5	75.4	0.229
μ	TFG	28.8 ± 1.6	24.5 ± 1.0	62.8	62.7	0.122
μ	BDG , $\tau_m = 0.5$	11.2 ± 1.1	35.1 ± 1.1	71.4	71.2	0.119
μ	BDG , $\tau_m = 1$	8.9 ± 0.2	40.5 ± 0.8	76.5	76.3	0.119
α	Unguided	5.0 ± 0.3	39.7 ± 0.6	75.6	75.5	0.050
α	DPS-style plug-in	6.1 ± 0.3	37.2 ± 1.6	73.7	73.6	0.118
α	TMPD-inspired	6.1 ± 0.4	38.9 ± 1.1	75.1	75.0	0.200
α	LGD-MC	5.2 ± 0.4	39.2 ± 1.3	74.9	74.9	0.224
α	TFG	10.5 ± 0.3	26.8 ± 0.8	61.2	61.1	0.121
α	BDG , $\tau_m = 0.5$	7.0 ± 0.5	36.4 ± 1.0	72.2	72.1	0.118
α	BDG , $\tau_m = 1$	5.6 ± 0.5	37.9 ± 1.1	75.1	75.1	0.118
gap	Unguided	10.3 ± 1.0	39.7 ± 0.6	75.6	75.5	0.051
gap	DPS-style plug-in	12.6 ± 0.7	33.6 ± 0.8	69.9	69.8	0.119
gap	TMPD-inspired	11.6 ± 0.9	37.3 ± 0.7	73.8	73.7	0.201
gap	LGD-MC	10.8 ± 0.7	39.5 ± 0.8	75.2	75.1	0.225
gap	TFG	27.5 ± 0.3	14.4 ± 0.8	45.5	45.5	0.122
gap	BDG , $\tau_m = 0.5$	14.2 ± 0.8	30.0 ± 0.4	65.8	65.7	0.119
gap	BDG , $\tau_m = 1$	10.6 ± 0.2	37.6 ± 0.8	74.3	74.2	0.119

Table S5: Complete headline evaluation on borrowed EquiFM. IB is decoded; IB and stability show percent mean $\pm$ seed sd. Validity and distinct-valid yield (DV) are percentages; time is recorded cell wall-clock divided by attempts. $w = 1$, $t \geq 0.5$, three seeds, $n = 2,000$ each.

Property	Method	IB $\uparrow$	Stable $\uparrow$	Valid $\uparrow$	DV $\uparrow$	s/sample
μ	Unguided	8.7 ± 0.8	86.2 ± 0.3	93.8	93.5	0.086
μ	DPS-style plug-in	10.9 ± 0.9	82.1 ± 0.7	91.9	91.6	0.231
μ	TMPD-inspired	9.0 ± 0.3	85.3 ± 0.4	93.5	93.2	0.366
μ	LGD-MC	9.7 ± 0.3	85.5 ± 0.4	93.3	93.1	0.474
μ	TFG	11.9 ± 0.3	74.0 ± 0.7	86.5	86.2	0.301
μ	BDG , $\tau_m = 0.5$	12.1 ± 0.5	79.3 ± 0.7	90.9	90.6	0.232
μ	BDG , $\tau_m = 1$	8.9 ± 0.7	85.8 ± 0.3	93.6	93.3	0.232
α	Unguided	2.1 ± 0.1	86.2 ± 0.3	93.8	93.5	0.087
α	DPS-style plug-in	2.1 ± 0.2	82.9 ± 0.4	91.9	91.7	0.233
α	TMPD-inspired	2.1 ± 0.3	84.8 ± 0.3	93.3	93.0	0.368
α	LGD-MC	2.1 ± 0.1	85.5 ± 0.9	93.3	93.0	0.476
α	TFG	3.4 ± 0.5	64.5 ± 0.5	84.5	84.4	0.302
α	BDG , $\tau_m = 0.5$	2.5 ± 0.2	76.6 ± 0.5	89.5	89.2	0.233
α	BDG , $\tau_m = 1$	2.0 ± 0.1	84.8 ± 0.5	92.8	92.6	0.233
gap	Unguided	4.9 ± 0.2	86.2 ± 0.3	93.8	93.5	0.086
gap	DPS-style plug-in	5.9 ± 0.3	82.7 ± 0.4	91.9	91.7	0.231
gap	TMPD-inspired	5.1 ± 0.3	84.6 ± 0.7	92.7	92.4	0.365
gap	LGD-MC	5.1 ± 0.4	86.1 ± 0.2	93.5	93.3	0.473
gap	TFG	10.9 ± 0.2	55.5 ± 1.4	76.4	76.2	0.300
gap	BDG , $\tau_m = 0.5$	6.8 ± 0.8	77.4 ± 1.1	90.1	90.0	0.231
gap	BDG , $\tau_m = 1$	5.0 ± 0.4	85.6 ± 0.3	93.4	93.1	0.231

Table S6: Complete headline evaluation on borrowed EDMsecond. IB is decoded; IB and stability show percent mean $\pm$ seed sd. Validity and distinct-valid yield (DV) are percentages; time is recorded cell wall-clock divided by attempts. $w = 1$, $t \geq 0.5$, three seeds, $n = 2,000$ each.

Property	Method	IB $\uparrow$	Stable $\uparrow$	Valid $\uparrow$	DV $\uparrow$	s/sample
μ	Unguided	9.4 ± 0.2	67.2 ± 1.3	86.0	85.8	0.087
μ	DPS-style plug-in	11.2 ± 1.1	62.2 ± 0.6	82.8	82.7	0.210
α	Unguided	5.4 ± 0.7	67.2 ± 1.2	86.0	85.8	0.087
α	DPS-style plug-in	6.1 ± 0.4	64.6 ± 0.8	84.1	83.9	0.208
gap	Unguided	10.8 ± 0.7	67.1 ± 1.3	86.0	85.8	0.087
gap	DPS-style plug-in	12.8 ± 0.7	57.9 ± 1.7	80.3	80.2	0.209

Table S7: Complete headline evaluation on our VP diffusion. IB is decoded; IB and stability show percent mean $\pm$ seed sd. Validity and distinct-valid yield (DV) are percentages; time is recorded cell wall-clock divided by attempts. $w = 1$, $t \geq 0.5$, three seeds, $n = 2,000$ each.

Property	Method	IB $\uparrow$	Stable $\uparrow$	Valid $\uparrow$	DV $\uparrow$	s/sample
μ	Unguided	8.7 ± 0.8	28.8 ± 1.3	66.2	66.1	0.048
μ	DPS-style plug-in	10.8 ± 0.4	26.2 ± 0.9	61.8	61.7	0.114
α	Unguided	5.2 ± 0.6	28.8 ± 1.3	66.2	66.1	0.048
α	DPS-style plug-in	5.4 ± 0.4	25.2 ± 1.5	58.7	58.6	0.114
gap	Unguided	9.8 ± 0.6	28.8 ± 1.3	66.2	66.1	0.048
gap	DPS-style plug-in	12.4 ± 1.2	23.2 ± 0.6	57.3	57.3	0.114

Table S8: BDG minus plug-in on decoded molecular IB. Differences, paired seed sd and the threshold-sized difference $2.99 \text{ se}(\Delta)$ are in percentage points. The registered unpaired statistic is shown separately from descriptive pairing.

Backend	Property	τ_m	ΔIB	$\text{SD}(\Delta)$	z	Threshold
fm	μ	0.5	+0.63	1.45	+1.11	1.70
fm	μ	1	-1.68	0.19	-3.11	1.62
fm	α	0.5	+0.90	0.50	+2.00	1.35
fm	α	1	-0.43	0.75	-1.01	1.28
fm	gap	0.5	+1.52	0.21	+2.44	1.86
fm	gap	1	-2.07	0.72	-3.54	1.75
equifm	μ	0.5	+1.25	0.70	+2.15	1.74
equifm	μ	1	-1.93	0.62	-3.55	1.63
equifm	α	0.5	+0.45	0.13	+1.65	0.81
equifm	α	1	-0.02	0.31	-0.06	0.77
equifm	gap	0.5	+0.90	0.56	+2.02	1.33
equifm	gap	1	-0.93	0.67	-2.25	1.24

A.6 Complete Molecular Ablation Grids

Figures S2 and S3 show every nonzero gain/setpoint combination as decoded coverage change from its own zero-gain control, with strengths kept separate. Tables S10, S11, S12, S13, S14 and S15 provide the complete numeric ablations: 204 configurations, each evaluated with three seeds, accounting for all 612 cells. Each caption specifies its property, q50 target and band. These tables extend the selected dipole rows in Table 2 with both strengths, all gains and setpoints, target error, bias, spread and stability.

For our FM dipole task at $w = 1$, zero gain gives MAE 0.969 D and pooled mean 2.615 D around $y = 2.4932$ D. At $\tau_m = 0.5$, gains 4 and 8 lower MAE to 0.903 and 0.861 D and move the mean to 2.609 and 2.603 D. This supports improved endpoint target concentration in the selected ablation. Mean bias is not uniformly improved across properties: our FM gap bias grows slightly in magnitude at gain 4 despite lower MAE. Increasing gain also need not monotonically increase coverage, especially at $w = 4$. The wider setpoint lowers occupancy of the fixed band in every grid, illustrating the intended spread-occupancy trade-off.

At $\eta = 8$, $w = 1$, $\tau_m = 1.5$, decoded spread exceeds unguided for all three properties and both flow backends, approximately 1.15–1.25 times the reference. The tested positive plug-in strengths $w \in \{1, 4\}$ remain below it. This comparison does not exclude signed strengths, arbitrary schedules, or a shifted plug-in target from reaching similar states. One-sided residuals, open-loop replay and signed-strength controls were not part of the completed grid.

Across the four setpoints at $\eta = 8$, $w = 1$, all six decoded spread curves increase monotonically (Figure 2a). The step-averaged deviation coefficient and decoded coverage have Spearman correlations of 0.92–0.99 across the 16 nonzero configurations in each of the 12 fixed-backbone/property/strength grids. This descriptive association links coefficient changes to occupancy; it does not separate closed-loop feedback from guidance magnitude.

Table S9 compares $\Delta\text{IB}/(-\Delta\text{stability})$, with both differences measured in percentage points from the same ablation-stage plug-in reference at $w = 1$. Contracting BDG at $\eta = 8$ gains more coverage per stability point lost than increasing plug-in strength to $w = 4$ in five of six tasks; at the headline gain $\eta = 4$, this holds in four of six. These exploratory ratios are sensitive to small denominators, do not establish statistical superiority, and are not joint in-band-and-stable yield. The full grid reports the underlying metrics.

Table S9: Exploratory coverage gained per stability point lost. All differences use the ablation-stage zero-gain reference at $w = 1$. BDG changes η at $\tau_m = 0.5, w = 1$; the strength control keeps $\eta = 0$ and increases w to 4. Ratios use decoded IB and molecular stability means, not joint useful yield.

Backend	Property	BDG $\eta = 4$	BDG $\eta = 8$	Plug-in $w = 4$
Our FM	μ	1.267	0.792	0.534
Our FM	α	1.174	0.806	0.439
Our FM	gap	0.423	0.524	0.383
EquiFM	μ	0.370	0.385	0.369
EquiFM	α	0.053	0.057	0.083
EquiFM	gap	0.149	0.238	0.217

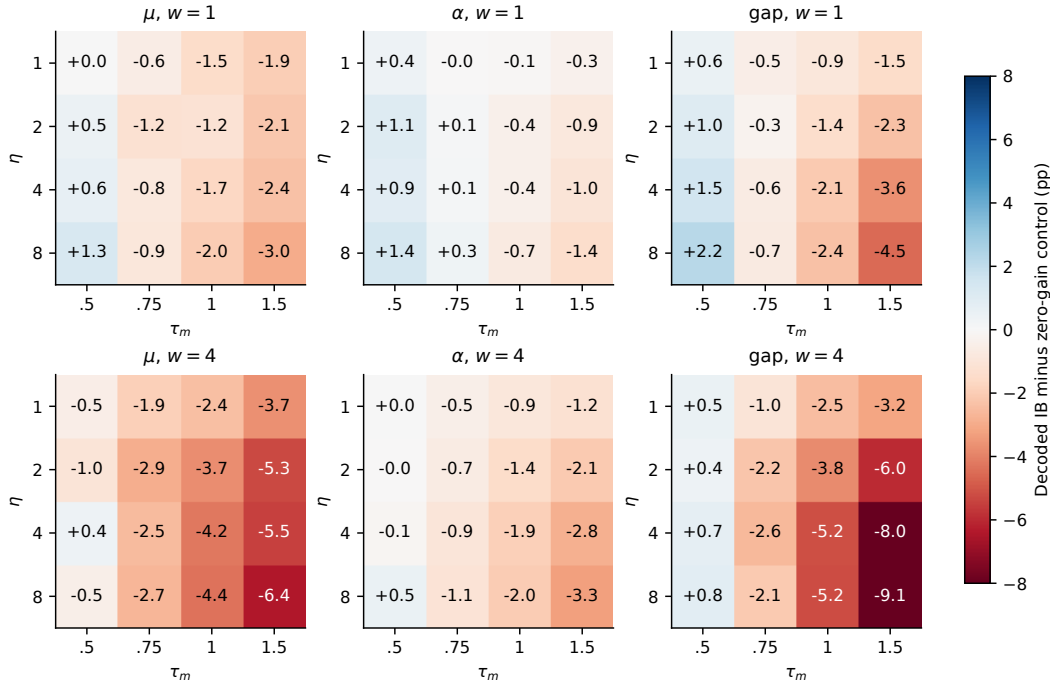


Figure S2: Our FM: full decoded coverage ablation. Each panel fixes property and strength; entries are percentage-point changes from $\eta = 0$ at that strength. The color scale is shared with Figure S3; values outside its range retain their exact text.

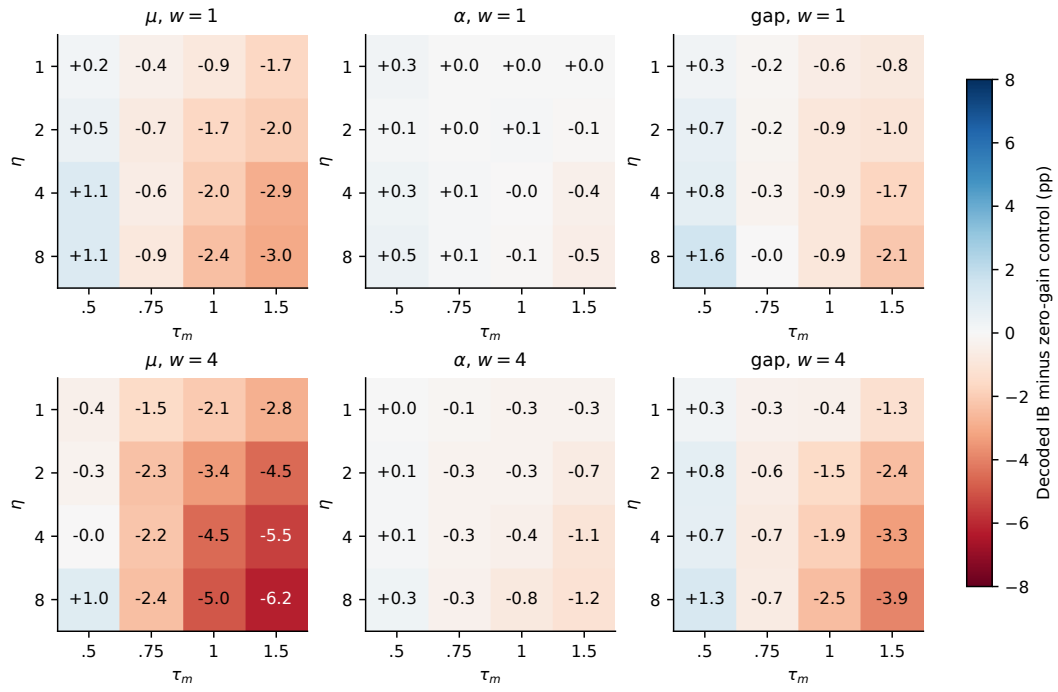


Figure S3: Borrowed EquiFM: the same complete grid and matched zero-gain contrasts. Rerun variability accompanies the three-seed averages.

Table S10: Our FM, μ : all gain/setpoint ablations at $w = 1, 4$, q50 target $y = 2.4932$ D, half-width $\delta = 0.17541$ D. Each row uses three seeds of 2,000 samples. IB and molecular stability are percent mean $\pm$ seed sd. MAE, signed bias of the pooled mean, and pooled standard deviation σ are decoded and normalized by δ . The $\eta = 0$ row removes the residual; its setpoint is irrelevant.

w	η	τ_m	IB $\uparrow$	MAE/ δ $\downarrow$	Bias/ δ	σ/δ	Stable $\uparrow$
1	0	–	10.6 $\pm$ 0.4	5.53	+0.69	6.89	35.6 $\pm$ 0.3
1	1	0.5	10.6 $\pm$ 0.2	5.42	+0.75	6.74	35.7 $\pm$ 0.2
1	1	0.75	10.0 $\pm$ 0.0	5.77	+0.89	7.13	37.8 $\pm$ 0.5
1	1	1	9.1 $\pm$ 0.1	6.00	+1.03	7.45	38.9 $\pm$ 1.2
1	1	1.5	8.6 $\pm$ 0.2	6.24	+1.21	7.81	39.2 $\pm$ 0.7
1	2	0.5	11.1 $\pm$ 0.5	5.30	+0.70	6.56	35.8 $\pm$ 0.5
1	2	0.75	9.4 $\pm$ 0.3	5.90	+1.01	7.33	38.8 $\pm$ 1.2
1	2	1	9.4 $\pm$ 0.4	6.23	+1.25	7.77	39.6 $\pm$ 0.4
1	2	1.5	8.5 $\pm$ 0.5	6.81	+1.45	8.82	39.4 $\pm$ 0.9
1	4	0.5	11.2 $\pm$ 1.1	5.15	+0.66	6.35	35.1 $\pm$ 1.1
1	4	0.75	9.8 $\pm$ 0.5	5.91	+1.03	7.30	38.9 $\pm$ 0.3
1	4	1	8.9 $\pm$ 0.2	6.46	+1.26	8.17	40.5 $\pm$ 0.8
1	4	1.5	8.2 $\pm$ 0.7	7.27	+1.71	9.93	38.2 $\pm$ 1.0
1	8	0.5	11.9 $\pm$ 0.3	4.91	+0.62	6.07	34.0 $\pm$ 0.5
1	8	0.75	9.7 $\pm$ 0.4	5.82	+1.02	7.21	38.7 $\pm$ 0.5
1	8	1	8.6 $\pm$ 0.4	6.64	+1.37	8.36	40.4 $\pm$ 0.7
1	8	1.5	7.6 $\pm$ 0.2	7.85	+2.08	10.96	37.6 $\pm$ 1.2
4	0	–	13.4 $\pm$ 0.4	4.63	+0.31	5.84	30.3 $\pm$ 0.6
4	1	0.5	12.9 $\pm$ 0.5	4.70	+0.27	5.85	31.2 $\pm$ 0.5
4	1	0.75	11.6 $\pm$ 0.5	5.10	+0.44	6.31	34.3 $\pm$ 0.4
4	1	1	11.0 $\pm$ 0.3	5.33	+0.58	6.60	35.4 $\pm$ 0.7
4	1	1.5	9.8 $\pm$ 0.2	5.76	+0.72	7.15	38.1 $\pm$ 1.4
4	2	0.5	12.4 $\pm$ 0.8	4.64	+0.28	5.72	31.8 $\pm$ 0.3
4	2	0.75	10.5 $\pm$ 0.7	5.40	+0.62	6.71	36.0 $\pm$ 1.1
4	2	1	9.7 $\pm$ 0.4	5.88	+0.75	7.30	39.1 $\pm$ 0.9
4	2	1.5	8.1 $\pm$ 0.8	6.76	+1.09	8.79	38.6 $\pm$ 1.2
4	4	0.5	13.8 $\pm$ 0.4	4.50	+0.34	5.58	33.5 $\pm$ 0.6
4	4	0.75	10.9 $\pm$ 0.6	5.40	+0.58	6.70	37.4 $\pm$ 1.0
4	4	1	9.2 $\pm$ 0.6	6.28	+0.82	7.83	39.9 $\pm$ 0.2
4	4	1.5	8.0 $\pm$ 0.3	7.61	+1.40	10.44	36.6 $\pm$ 1.0
4	8	0.5	13.0 $\pm$ 0.5	4.46	+0.49	5.54	34.0 $\pm$ 1.6
4	8	0.75	10.8 $\pm$ 0.3	5.42	+1.01	6.81	37.7 $\pm$ 0.5
4	8	1	9.1 $\pm$ 0.8	6.41	+1.01	8.09	40.3 $\pm$ 0.7
4	8	1.5	7.0 $\pm$ 0.2	8.05	+1.55	11.08	37.5 $\pm$ 0.6

Table S11: Our FM, α : all gain/setpoint ablations at $w = 1, 4$, q50 target $y = 75.5400 \text{ Bohr}^3$, half-width $\delta = 0.50618 \text{ Bohr}^3$. Each row uses three seeds of 2,000 samples. IB and molecular stability are percent mean $\pm$ seed sd. MAE, signed bias of the pooled mean, and pooled standard deviation σ are decoded and normalized by δ . The $\eta = 0$ row removes the residual; its setpoint is irrelevant.

w	η	τ_m	IB $\uparrow$	MAE/ $\delta \downarrow$	Bias/ δ	σ/δ	Stable $\uparrow$
1	0	—	6.1 ± 0.3	10.47	+1.79	13.23	37.2 ± 1.6
1	1	0.5	6.5 ± 0.1	9.65	+1.74	12.30	37.7 ± 1.8
1	1	0.75	6.1 ± 0.2	10.53	+1.81	13.30	37.8 ± 1.6
1	1	1	6.0 ± 0.0	11.12	+1.69	14.05	37.8 ± 1.3
1	1	1.5	5.8 ± 0.4	11.74	+1.64	14.91	38.3 ± 1.0
1	2	0.5	7.1 ± 0.6	9.29	+1.77	11.90	37.0 ± 1.1
1	2	0.75	6.2 ± 0.4	10.56	+1.82	13.34	37.8 ± 1.3
1	2	1	5.7 ± 0.3	11.59	+1.71	14.67	38.5 ± 0.5
1	2	1.5	5.2 ± 0.2	12.72	+1.16	16.39	38.3 ± 0.7
1	4	0.5	7.0 ± 0.5	9.00	+1.78	11.56	36.4 ± 1.0
1	4	0.75	6.2 ± 0.2	10.56	+1.80	13.34	37.6 ± 1.2
1	4	1	5.6 ± 0.5	11.99	+1.58	15.24	37.9 ± 1.1
1	4	1.5	5.1 ± 0.4	13.61	+0.57	17.81	37.8 ± 0.4
1	8	0.5	7.5 ± 0.4	8.67	+1.75	11.16	35.5 ± 0.3
1	8	0.75	6.4 ± 0.1	10.52	+1.81	13.31	37.9 ± 1.7
1	8	1	5.4 ± 0.3	12.39	+1.40	15.81	38.2 ± 0.5
1	8	1.5	4.7 ± 0.4	14.69	+0.04	19.38	37.3 ± 0.3
4	0	—	7.5 ± 0.4	8.58	+2.85	10.81	34.1 ± 1.1
4	1	0.5	7.5 ± 0.2	8.53	+2.93	10.71	33.2 ± 1.0
4	1	0.75	7.0 ± 0.1	9.32	+2.99	11.66	35.9 ± 0.1
4	1	1	6.6 ± 0.4	9.89	+3.06	12.31	36.4 ± 1.1
4	1	1.5	6.3 ± 0.4	10.71	+2.94	13.35	37.6 ± 0.8
4	2	0.5	7.4 ± 0.1	8.49	+2.97	10.61	33.4 ± 1.0
4	2	0.75	6.8 ± 0.2	9.75	+3.12	12.12	36.5 ± 0.4
4	2	1	6.0 ± 0.5	10.97	+2.99	13.66	37.9 ± 0.7
4	2	1.5	5.4 ± 0.4	12.74	+2.22	16.31	38.2 ± 0.4
4	4	0.5	7.3 ± 0.1	8.38	+2.94	10.42	33.5 ± 0.9
4	4	0.75	6.5 ± 0.4	10.10	+3.09	12.52	36.8 ± 0.8
4	4	1	5.6 ± 0.3	11.80	+2.77	14.86	38.1 ± 1.0
4	4	1.5	4.6 ± 0.3	14.33	+1.20	18.70	36.8 ± 1.3
4	8	0.5	8.0 ± 0.5	8.21	+2.93	10.23	33.1 ± 0.1
4	8	0.75	6.4 ± 0.4	10.27	+3.08	12.73	37.2 ± 0.9
4	8	1	5.4 ± 0.1	12.26	+2.51	15.68	38.8 ± 0.9
4	8	1.5	4.2 ± 0.3	16.11	+0.49	20.85	35.8 ± 0.8

Table S12: Our FM, gap: all gain/setpoint ablations at $w = 1, 4$, q50 target $y = 0.2496$ Ha, half-width $\delta = 0.00748$ Ha. Each row uses three seeds of 2,000 samples. IB and molecular stability are percent mean $\pm$ seed sd. MAE, signed bias of the pooled mean, and pooled standard deviation σ are decoded and normalized by δ . The $\eta = 0$ row removes the residual; its setpoint is irrelevant.

w	η	τ_m	IB $\uparrow$	MAE/ δ $\downarrow$	Bias/ δ	σ/δ	Stable $\uparrow$
1	0	—	12.6 $\pm$ 0.7	4.51	-1.26	5.29	33.6 $\pm$ 0.8
1	1	0.5	13.2 $\pm$ 0.7	4.36	-1.30	5.12	31.3 $\pm$ 0.6
1	1	0.75	12.1 $\pm$ 0.8	4.57	-1.25	5.37	34.1 $\pm$ 0.3
1	1	1	11.8 $\pm$ 1.1	4.74	-1.26	5.56	36.6 $\pm$ 0.6
1	1	1.5	11.2 $\pm$ 0.7	4.95	-1.18	5.81	37.6 $\pm$ 0.1
1	2	0.5	13.7 $\pm$ 0.5	4.24	-1.37	4.95	31.0 $\pm$ 0.9
1	2	0.75	12.4 $\pm$ 0.9	4.60	-1.29	5.40	35.0 $\pm$ 1.3
1	2	1	11.3 $\pm$ 0.8	4.90	-1.21	5.75	37.4 $\pm$ 0.9
1	2	1.5	10.3 $\pm$ 0.4	5.31	-1.15	6.24	38.2 $\pm$ 0.7
1	4	0.5	14.2 $\pm$ 0.8	4.13	-1.36	4.82	30.0 $\pm$ 0.4
1	4	0.75	12.0 $\pm$ 0.4	4.62	-1.31	5.40	35.1 $\pm$ 1.0
1	4	1	10.6 $\pm$ 0.2	5.09	-1.17	5.95	37.6 $\pm$ 0.8
1	4	1.5	9.0 $\pm$ 0.5	5.70	-1.01	6.69	35.3 $\pm$ 0.4
1	8	0.5	14.8 $\pm$ 0.8	3.96	-1.28	4.64	29.3 $\pm$ 0.5
1	8	0.75	11.9 $\pm$ 0.2	4.58	-1.19	5.38	35.4 $\pm$ 0.7
1	8	1	10.3 $\pm$ 0.4	5.16	-1.19	6.05	37.4 $\pm$ 0.6
1	8	1.5	8.1 $\pm$ 0.5	6.01	-0.87	7.02	33.0 $\pm$ 1.5
4	0	—	15.6 $\pm$ 0.8	3.89	-1.14	4.63	25.8 $\pm$ 0.8
4	1	0.5	16.1 $\pm$ 0.7	3.84	-1.23	4.55	25.4 $\pm$ 0.4
4	1	0.75	14.6 $\pm$ 0.5	4.11	-1.24	4.85	28.8 $\pm$ 1.1
4	1	1	13.1 $\pm$ 0.4	4.36	-1.23	5.13	30.8 $\pm$ 1.0
4	1	1.5	12.5 $\pm$ 0.1	4.60	-1.26	5.39	33.7 $\pm$ 1.0
4	2	0.5	16.0 $\pm$ 0.4	3.80	-1.23	4.49	25.4 $\pm$ 0.7
4	2	0.75	13.4 $\pm$ 0.7	4.28	-1.25	5.05	30.2 $\pm$ 0.7
4	2	1	11.9 $\pm$ 0.2	4.65	-1.19	5.47	34.9 $\pm$ 0.6
4	2	1.5	9.6 $\pm$ 0.3	5.40	-1.10	6.34	36.5 $\pm$ 0.7
4	4	0.5	16.4 $\pm$ 0.4	3.74	-1.20	4.42	27.0 $\pm$ 0.3
4	4	0.75	13.0 $\pm$ 0.4	4.35	-1.14	5.14	33.0 $\pm$ 1.4
4	4	1	10.5 $\pm$ 0.8	4.96	-1.08	5.83	36.6 $\pm$ 0.6
4	4	1.5	7.6 $\pm$ 0.6	6.06	-0.90	7.05	33.7 $\pm$ 1.1
4	8	0.5	16.5 $\pm$ 0.4	3.67	-0.97	4.41	27.0 $\pm$ 0.6
4	8	0.75	13.5 $\pm$ 1.1	4.39	-0.96	5.24	34.1 $\pm$ 0.3
4	8	1	10.4 $\pm$ 0.2	5.08	-1.05	5.97	38.1 $\pm$ 0.6
4	8	1.5	6.5 $\pm$ 0.6	6.47	-0.77	7.47	31.5 $\pm$ 1.6

Table S13: Pretrained EquiFM (flow matching), μ : all gain/setpoint ablations at $w = 1, 4$, q50 target $y = 2.4932$ D, half-width $\delta = 0.15675$ D. Each row uses three seeds of 2,000 samples. IB and molecular stability are percent mean $\pm$ seed sd. MAE, signed bias of the pooled mean, and pooled standard deviation σ are decoded and normalized by δ . The $\eta = 0$ row removes the residual; its setpoint is irrelevant.

w	η	τ_m	IB $\uparrow$	MAE/ δ $\downarrow$	Bias/ δ	σ/δ	Stable $\uparrow$
1	0	–	10.9 $\pm$ 0.9	5.71	+1.27	7.17	81.9 $\pm$ 0.9
1	1	0.5	11.1 $\pm$ 0.9	5.45	+1.03	6.79	80.9 $\pm$ 0.5
1	1	0.75	10.5 $\pm$ 1.0	5.84	+1.28	7.28	82.6 $\pm$ 1.1
1	1	1	10.1 $\pm$ 1.0	6.09	+1.46	7.59	83.6 $\pm$ 1.2
1	1	1.5	9.2 $\pm$ 0.9	6.41	+1.74	7.99	85.0 $\pm$ 0.9
1	2	0.5	11.4 $\pm$ 0.5	5.38	+1.02	6.73	80.0 $\pm$ 0.9
1	2	0.75	10.2 $\pm$ 0.7	5.97	+1.40	7.48	83.5 $\pm$ 1.2
1	2	1	9.2 $\pm$ 0.2	6.42	+1.72	7.97	84.9 $\pm$ 0.5
1	2	1.5	9.0 $\pm$ 0.4	7.11	+2.03	9.68	85.4 $\pm$ 0.6
1	4	0.5	12.0 $\pm$ 0.3	5.22	+0.85	6.58	79.0 $\pm$ 0.2
1	4	0.75	10.3 $\pm$ 0.1	6.00	+1.39	7.50	84.1 $\pm$ 1.0
1	4	1	8.9 $\pm$ 0.7	6.70	+1.83	8.52	85.5 $\pm$ 0.3
1	4	1.5	8.0 $\pm$ 0.1	7.60	+2.21	10.50	83.0 $\pm$ 0.9
1	8	0.5	12.0 $\pm$ 0.2	5.08	+0.84	6.31	79.0 $\pm$ 0.9
1	8	0.75	10.1 $\pm$ 0.6	6.01	+1.36	7.43	84.7 $\pm$ 0.5
1	8	1	8.5 $\pm$ 0.7	6.96	+2.00	8.87	85.4 $\pm$ 0.6
1	8	1.5	7.9 $\pm$ 0.3	8.02	+2.41	11.38	82.0 $\pm$ 0.7
4	0	–	13.2 $\pm$ 0.5	4.77	+0.59	6.07	75.6 $\pm$ 0.6
4	1	0.5	12.8 $\pm$ 0.7	4.79	+0.68	6.00	75.9 $\pm$ 0.7
4	1	0.75	11.7 $\pm$ 0.7	5.21	+0.86	6.54	78.7 $\pm$ 0.8
4	1	1	11.1 $\pm$ 1.2	5.46	+0.96	6.82	80.9 $\pm$ 0.7
4	1	1.5	10.4 $\pm$ 0.9	5.77	+1.08	7.24	83.1 $\pm$ 0.7
4	2	0.5	12.9 $\pm$ 1.0	4.78	+0.64	6.01	76.1 $\pm$ 1.3
4	2	0.75	10.9 $\pm$ 0.6	5.46	+0.98	6.82	81.3 $\pm$ 0.3
4	2	1	9.8 $\pm$ 0.4	6.00	+1.21	7.41	84.8 $\pm$ 0.6
4	2	1.5	8.7 $\pm$ 0.9	7.06	+1.59	9.34	84.8 $\pm$ 1.0
4	4	0.5	13.2 $\pm$ 0.7	4.70	+0.55	5.88	76.5 $\pm$ 0.4
4	4	0.75	11.0 $\pm$ 0.7	5.62	+1.07	7.03	83.3 $\pm$ 0.7
4	4	1	8.7 $\pm$ 0.5	6.52	+1.33	8.16	85.5 $\pm$ 0.8
4	4	1.5	7.8 $\pm$ 0.2	7.84	+1.81	10.95	81.7 $\pm$ 0.7
4	8	0.5	14.2 $\pm$ 0.2	4.60	+0.49	5.87	78.5 $\pm$ 0.4
4	8	0.75	10.8 $\pm$ 0.4	5.69	+1.13	7.17	83.4 $\pm$ 0.7
4	8	1	8.2 $\pm$ 0.5	6.87	+1.58	8.72	85.2 $\pm$ 0.8
4	8	1.5	7.0 $\pm$ 0.3	8.25	+1.93	11.54	81.2 $\pm$ 0.6

Table S14: Pretrained EquiFM (flow matching), α : all gain/setpoint ablations at $w = 1, 4$, q50 target $y = 75.5400$ Bohr³, half-width $\delta = 0.16480$ Bohr³. Each row uses three seeds of 2,000 samples. IB and molecular stability are percent mean $\pm$ seed sd. MAE, signed bias of the pooled mean, and pooled standard deviation σ are decoded and normalized by δ . The $\eta = 0$ row removes the residual; its setpoint is irrelevant.

w	η	τ_m	IB $\uparrow$	MAE/ δ $\downarrow$	Bias/ δ	σ/δ	Stable $\uparrow$
1	0	—	2.1 ± 0.2	32.49	-11.31	41.35	82.8 ± 0.4
1	1	0.5	2.3 ± 0.3	30.38	-11.58	38.68	80.5 ± 1.0
1	1	0.75	2.1 ± 0.2	32.16	-11.25	40.89	82.7 ± 0.6
1	1	1	2.1 ± 0.1	33.29	-10.91	42.32	83.9 ± 1.0
1	1	1.5	2.1 ± 0.2	34.74	-11.02	44.12	85.0 ± 0.6
1	2	0.5	2.2 ± 0.4	29.62	-11.95	37.43	78.9 ± 1.1
1	2	0.75	2.1 ± 0.3	31.97	-11.30	40.62	82.7 ± 0.1
1	2	1	2.1 ± 0.1	34.28	-10.96	43.51	84.7 ± 0.5
1	2	1.5	2.0 ± 0.3	37.81	-12.09	48.48	85.2 ± 0.5
1	4	0.5	2.4 ± 0.3	28.49	-11.89	35.97	76.8 ± 0.2
1	4	0.75	2.2 ± 0.2	31.86	-11.40	40.53	82.6 ± 0.2
1	4	1	2.0 ± 0.1	35.53	-11.10	45.22	84.8 ± 0.5
1	4	1.5	1.7 ± 0.3	42.28	-13.62	54.00	83.1 ± 0.8
1	8	0.5	2.5 ± 0.1	27.31	-11.93	34.50	74.6 ± 0.6
1	8	0.75	2.1 ± 0.1	31.53	-11.46	40.22	82.0 ± 0.5
1	8	1	2.0 ± 0.2	36.79	-11.39	47.03	85.1 ± 0.2
1	8	1.5	1.5 ± 0.4	46.61	-15.04	58.60	79.3 ± 0.5
4	0	—	2.6 ± 0.1	27.68	-9.36	36.11	76.2 ± 0.4
4	1	0.5	2.6 ± 0.3	26.55	-10.41	34.43	74.5 ± 0.5
4	1	0.75	2.5 ± 0.2	28.25	-9.39	36.83	77.2 ± 0.9
4	1	1	2.3 ± 0.1	29.85	-9.19	38.77	79.7 ± 0.6
4	1	1.5	2.3 ± 0.3	32.05	-8.68	41.31	81.7 ± 0.3
4	2	0.5	2.7 ± 0.1	25.84	-10.41	33.51	72.6 ± 0.6
4	2	0.75	2.3 ± 0.2	28.50	-9.40	37.20	77.6 ± 0.6
4	2	1	2.4 ± 0.3	32.10	-8.77	41.32	81.0 ± 0.6
4	2	1.5	2.0 ± 0.1	38.30	-9.75	49.83	83.2 ± 0.2
4	4	0.5	2.7 ± 0.3	24.96	-10.87	32.39	71.0 ± 0.9
4	4	0.75	2.4 ± 0.2	29.08	-9.49	37.85	78.4 ± 0.8
4	4	1	2.2 ± 0.3	34.67	-8.75	44.55	83.7 ± 0.7
4	4	1.5	1.6 ± 0.2	45.53	-12.35	58.23	78.7 ± 0.4
4	8	0.5	2.9 ± 0.1	24.27	-11.20	31.38	69.1 ± 0.7
4	8	0.75	2.3 ± 0.2	29.59	-9.44	38.31	78.3 ± 0.6
4	8	1	1.8 ± 0.2	36.45	-9.33	47.05	83.7 ± 0.6
4	8	1.5	1.4 ± 0.2	50.89	-13.74	63.38	72.0 ± 0.3

Table S15: Pretrained EquiFM (flow matching), gap; all gain/setpoint ablations at $w = 1, 4$, q50 target $y = 0.2496$ Ha, half-width $\delta = 0.00374$ Ha. Each row uses three seeds of 2,000 samples. IB and molecular stability are percent mean $\pm$ seed sd. MAE, signed bias of the pooled mean, and pooled standard deviation σ are decoded and normalized by δ . The $\eta = 0$ row removes the residual; its setpoint is irrelevant.

w	η	τ_m	IB $\uparrow$	MAE/ δ $\downarrow$	Bias/ δ	σ/δ	Stable $\uparrow$
1	0	—	5.9 ± 0.3	9.07	-1.16	10.63	82.7 ± 0.5
1	1	0.5	6.2 ± 0.7	8.59	-1.23	10.10	80.7 ± 0.6
1	1	0.75	5.7 ± 0.0	9.09	-1.13	10.66	82.5 ± 0.4
1	1	1	5.3 ± 0.2	9.42	-1.16	11.02	84.0 ± 0.1
1	1	1.5	5.1 ± 0.2	9.79	-1.22	11.43	85.0 ± 0.9
1	2	0.5	6.6 ± 0.5	8.45	-1.08	9.99	79.3 ± 0.4
1	2	0.75	5.6 ± 0.3	9.12	-1.11	10.69	82.5 ± 0.3
1	2	1	5.0 ± 0.1	9.76	-1.16	11.40	84.9 ± 0.7
1	2	1.5	4.9 ± 0.2	10.73	-1.39	12.52	85.4 ± 0.2
1	4	0.5	6.7 ± 0.6	8.21	-1.00	9.73	77.3 ± 1.5
1	4	0.75	5.6 ± 0.1	9.11	-0.97	10.68	82.9 ± 0.6
1	4	1	5.0 ± 0.4	10.16	-1.23	11.85	85.7 ± 0.3
1	4	1.5	4.1 ± 0.2	11.71	-1.31	13.58	80.7 ± 0.9
1	8	0.5	7.5 ± 0.4	7.88	-0.99	9.40	75.9 ± 1.0
1	8	0.75	5.9 ± 0.3	9.07	-0.91	10.66	82.9 ± 0.5
1	8	1	5.0 ± 0.2	10.48	-1.32	12.22	86.5 ± 0.7
1	8	1.5	3.7 ± 0.7	12.46	-1.43	14.34	74.8 ± 0.7
4	0	—	7.1 ± 0.4	8.03	-0.92	9.58	77.0 ± 1.1
4	1	0.5	7.4 ± 0.2	7.82	-0.91	9.34	75.0 ± 0.8
4	1	0.75	6.8 ± 0.8	8.30	-0.88	9.85	78.6 ± 0.7
4	1	1	6.7 ± 0.6	8.56	-1.05	10.10	80.2 ± 0.9
4	1	1.5	5.8 ± 0.3	9.07	-1.00	10.67	82.7 ± 0.6
4	2	0.5	7.9 ± 0.5	7.70	-0.99	9.22	74.7 ± 1.1
4	2	0.75	6.5 ± 0.8	8.44	-0.96	9.99	79.0 ± 1.2
4	2	1	5.6 ± 0.4	9.23	-0.84	10.84	83.5 ± 0.7
4	2	1.5	4.7 ± 0.0	10.99	-1.18	12.82	84.9 ± 0.4
4	4	0.5	7.8 ± 0.3	7.50	-0.75	9.02	73.2 ± 1.6
4	4	0.75	6.4 ± 0.6	8.57	-0.83	10.15	80.3 ± 1.4
4	4	1	5.2 ± 0.3	9.95	-1.01	11.64	84.9 ± 0.5
4	4	1.5	3.8 ± 0.7	12.41	-1.34	14.30	75.1 ± 0.1
4	8	0.5	8.4 ± 0.5	7.32	-0.70	8.87	72.1 ± 0.6
4	8	0.75	6.4 ± 0.2	8.72	-0.71	10.32	81.1 ± 0.9
4	8	1	4.6 ± 0.2	10.38	-1.07	12.12	86.2 ± 0.4
4	8	1.5	3.2 ± 0.7	13.22	-1.59	15.10	68.3 ± 1.1

A.7 Sequence Transfer and Failure Diagnostics

The checkpoint joins uniform Dirichlet noise to one-hot 500-base-pair sequences. At each step, the sampler clamps negatives, normalizes each position, and decodes final samples by the largest channel. Endpoint extrapolation for guidance can leave the simplex, so soft properties and decoded counts can disagree. For relaxed sequence p of length L ,

$$F_{GC}(p) = L^{-1} \sum_j (p_{j,C} + p_{j,G}), \quad F_{CpG}(p) = (L-1)^{-1} \sum_{j=1}^{L-1} p_{j,C} p_{j+1,G}.$$

GC is affine; CpG is quadratic. The final metric counts bases or adjacent pairs after decoding. Its half-width is $\delta = \max(0.16s, 4.4q)$, where the count quantum is $q = 1/500$ for GC and $q = 1/499$ for CpG. GC has $s = 0.055206$, $\delta/s = 0.160$; CpG has $s = 0.014748$, $\delta/s = 0.598$ because its lattice floor binds. Absolute coverage levels therefore cannot be ranked across properties or against molecular coverage.

Table S18 gives all recent-method adaptations, both registered BDG settings and both strengths. The 3-mer JSD reference is the training split. Hamming diversity uses the first 256 decoded sequences and all ordered pairs including self-pairs; it is almost unchanged around 0.7456 and is not evidence of functional diversity. Every fidelity number is paired with coverage at the same strength. All sequences decode to valid A/C/G/T strings, so categorical validity is uninformative here.

We checked the late-window comparison separately for GC and CpG, at $w = 1$ and 4, for every seed. Plug-in, TMPD and LGD-MC have exactly equal decoded coverage in all these cells. Across property means, standard deviations, JSD and Hamming diversity, their differences from plug-in are bounded by 4.22×10^{-6} for GC and 3.40×10^{-6} for CpG. TFG-MC remains distinct, with single-seed coverage differences from plug-in up to 0.05 points for GC and 0.75 for CpG. Thus the numerical overlap is specific to three adaptations under this window, not a claim that all methods or all DNA guidance settings coincide. Measured endpoint uncertainty declines about 38,000-fold before $t = 0.5$, consistent with suppression of the corrections that distinguish TMPD and LGD-MC. Dirichlet FM, Fisher Flow and MOG-DFM were not evaluated under these GC/CpG bands.

For GC in the original $t \geq 0.5$ window at $w = 4$, contracting BDG increases coverage while the normalized bias worsens from +0.997 to +1.231. Its clipping fraction rises to 9.24% from plug-in's 0.079%; CpG clipping is 7.41% versus 0.21%. Thus equal nominal strength does not equalize applied correction. A GC-only window diagnostic at $w = 1$ increased plug-in coverage by 22.82 points when guidance began at zero. BDG at $\tau_m = 0.5$ was 2.33 points below that plug-in result; $\tau_m = 1$ was 20.80 points below. About a third of early-window steps clip. The ratio of window gain to the best late-window GC gain is about 5.4, and no corresponding CpG early-window result was run.

A separate frozen DeepFlyBrain predictor diagnostic exists in `results/m2_dfb_activity.json`. Its port was not verified numerically against the published Keras model in the committed evidence, so we do not claim published-oracle equivalence or biological function from these scores. Base composition, low-order k-mers and Hamming diversity cannot establish enhancer activity or cell-type specificity.

New window follow-up. Commits `cfc5d64` and `f568f68` (28 September, 22:47) add the GC $t \geq 0.3$ comparison and diagnostic reports. At $w = 4$, plug-in/TMPD/LGD-MC/TFG-MC mean coverage is 30.25/29.18/29.97/29.77%; BDG ($\eta = 4, \tau_m = 0.5$) reaches 37.33%, a +7.08-point gain over plug-in (Table S16). Thus late-window numerical overlap is not an intrinsic GC or CpG property. This is an exploratory follow-up chosen after diagnosing endpoint uncertainty; it is separate from the original $t \geq 0.5$ protocol comparison. No three-seed CpG $t \geq 0.3$ or GC $\tau_m = 1$ follow-up was supplied.

At $t \geq 0.3$, GC JSD improves relative to unguided for all measured methods, although BDG's JSD remains higher than plug-in's. Clipping is about 20.8% for BDG versus 14.9% for plug-in. The $\eta = 0$ coefficient is algebraically identical to plug-in; same-seed reruns now differ numerically, and the implementation checks continuous metrics to 10^{-5} tolerance. The separate $n = 500$ window and strength diagnostics use different settings and do not replace the $n = 2000, w = 4$ comparison. In particular, the strength diagnostic identifies $w = 16$ at $t \geq 0.5$; it is not the strength used in Table 4.

Table S16: New GC window follow-up, $t \geq 0.3$, $w = 4$, three seeds of 2,000. IB is percent mean $\pm$ seed sd; differences and paired SD are percentage points against plug-in. JSD is $\times 10^{-4}$, Clip is percent of guided sample-steps. This exploratory follow-up was added after diagnosing the original late window.

Method	IB $\uparrow$	Δ IB	SD(Δ)	JSD $\downarrow$	Hamming	Clip	s/seq.
DPS-style plug-in	30.3 $\pm$ 2.0	+0.00	0.00	3.64	0.7456	14.94	0.124
TMPD-inspired	29.2 $\pm$ 2.1	-1.07	0.57	3.66	0.7455	14.89	0.145
LGD-MC	30.0 $\pm$ 1.3	-0.28	1.14	3.61	0.7455	15.13	0.131
TFG-MC adaptation	29.8 $\pm$ 1.8	-0.48	1.06	3.57	0.7455	15.17	0.109
BDG , $\tau_m = 0.5$	37.3 $\pm$ 6.5	+7.08	4.53	4.24	0.7455	20.75	0.122
BDG $\eta = 0$	30.3 $\pm$ 2.0	+0.00	0.00	3.63	0.7456	14.94	0.059

Table S17: DNA, $w = 4$, $t \geq 0.5$: decoded IB (percent mean $\pm$ seed sd); JSD ($\times 10^{-4}$) and Hamming diversity in GC/CpG order. Time is recorded s/sequence for CpG. TFG-MC is a restricted adaptation. Full results: Table S18.

Method	GC IB $\uparrow$	CpG IB $\uparrow$	JSD $\downarrow$	Hamming	Time
Unguided	14.0 $\pm$ 1.0	50.2 $\pm$ 1.6	4.53 / 4.53	0.7456 / 0.7456	0.049
DPS-style plug-in	14.1 $\pm$ 1.0	51.6 $\pm$ 1.4	4.61 / 4.63	0.7456 / 0.7456	0.096
TMPD-inspired	14.1 $\pm$ 1.0	51.6 $\pm$ 1.4	4.61 / 4.63	0.7456 / 0.7456	0.124
LGD-MC	14.1 $\pm$ 1.0	51.6 $\pm$ 1.4	4.61 / 4.63	0.7456 / 0.7456	0.102
TFG-MC adaptation	14.1 $\pm$ 1.0	51.1 $\pm$ 1.7	4.59 / 4.58	0.7456 / 0.7456	0.084
BDG , $\tau_m = 0.5$	16.4 $\pm$ 1.2	68.5 $\pm$ 1.2	5.93 / 7.73	0.7457 / 0.7457	0.076
BDG , $\tau_m = 1$	14.1 $\pm$ 1.0	50.3 $\pm$ 1.5	4.54 / 4.52	0.7456 / 0.7456	0.076

Table S18: All DNA headline settings. IB is percent mean $\pm$ seed sd; JSD is multiplied by 10^4 ; Conf. is decoding confidence; Div. is normalized Hamming distance; Clip is percent of guided sample-steps. Unguided rows repeat a common reference and are never pooled twice.

Prop., w	Method	IB $\uparrow$	JSD $\downarrow$	Conf.	Div.	Clip	s/seq.
GC, 1	Unguided	14.0 $\pm$ 1.0	4.53	0.958	0.7456	0.00	0.021
GC, 1	DPS-style plug-in	14.0 $\pm$ 0.9	4.53	0.958	0.7456	0.00	0.042
GC, 1	TMPD-inspired	14.0 $\pm$ 0.9	4.53	0.958	0.7456	0.00	0.045
GC, 1	LGD-MC	14.0 $\pm$ 0.9	4.53	0.958	0.7456	0.00	0.035
GC, 1	TFG-MC adaptation	14.1 $\pm$ 1.0	4.53	0.958	0.7456	0.00	0.047
GC, 1	BDG , $\tau_m = 0.5$	14.5 $\pm$ 1.0	4.95	0.956	0.7456	0.56	0.042
GC, 1	BDG , $\tau_m = 1$	14.0 $\pm$ 1.0	4.53	0.958	0.7456	0.00	0.042
GC, 4	Unguided	14.0 $\pm$ 1.0	4.53	0.958	0.7456	0.00	0.021
GC, 4	DPS-style plug-in	14.1 $\pm$ 1.0	4.61	0.958	0.7456	0.08	0.040
GC, 4	TMPD-inspired	14.1 $\pm$ 1.0	4.61	0.958	0.7456	0.08	0.051
GC, 4	LGD-MC	14.1 $\pm$ 1.0	4.61	0.958	0.7456	0.08	0.047
GC, 4	TFG-MC adaptation	14.1 $\pm$ 1.0	4.59	0.958	0.7456	0.06	0.043
GC, 4	BDG , $\tau_m = 0.5$	16.4 $\pm$ 1.2	5.93	0.947	0.7457	9.24	0.028
GC, 4	BDG , $\tau_m = 1$	14.1 $\pm$ 1.0	4.54	0.958	0.7456	0.01	0.028
CpG, 1	Unguided	50.2 $\pm$ 1.6	4.53	0.958	0.7456	0.00	0.036
CpG, 1	DPS-style plug-in	50.5 $\pm$ 1.8	4.53	0.958	0.7456	0.02	0.074
CpG, 1	TMPD-inspired	50.5 $\pm$ 1.8	4.53	0.958	0.7456	0.02	0.120
CpG, 1	LGD-MC	50.5 $\pm$ 1.8	4.53	0.958	0.7456	0.02	0.102
CpG, 1	TFG-MC adaptation	50.4 $\pm$ 1.7	4.54	0.958	0.7456	0.01	0.089
CpG, 1	BDG , $\tau_m = 0.5$	55.3 $\pm$ 0.8	5.26	0.956	0.7456	0.98	0.073
CpG, 1	BDG , $\tau_m = 1$	50.2 $\pm$ 1.6	4.53	0.958	0.7456	0.00	0.061
CpG, 4	Unguided	50.2 $\pm$ 1.6	4.53	0.958	0.7456	0.00	0.049
CpG, 4	DPS-style plug-in	51.6 $\pm$ 1.4	4.63	0.958	0.7456	0.21	0.096
CpG, 4	TMPD-inspired	51.6 $\pm$ 1.4	4.63	0.958	0.7456	0.21	0.124
CpG, 4	LGD-MC	51.6 $\pm$ 1.4	4.63	0.958	0.7456	0.21	0.102
CpG, 4	TFG-MC adaptation	51.1 $\pm$ 1.7	4.58	0.958	0.7456	0.10	0.084
CpG, 4	BDG , $\tau_m = 0.5$	68.5 $\pm$ 1.2	7.73	0.950	0.7457	7.41	0.076
CpG, 4	BDG , $\tau_m = 1$	50.3 $\pm$ 1.5	4.52	0.959	0.7456	0.03	0.076

A.8 Algorithm and Algebraic Limits

Algorithm 1 specifies the detached pullback. Setting $\eta = 0$ removes the added term exactly, giving the plug-in coefficient $y - F_i$ for every positive τ . Setting $\tau = 0$ is undefined because the residual divides by τ^2 ; the implementation rejects it. The zero-gain rows are therefore the valid plug-in ablation. For $B = 1$, the dispersion deviation vanishes. The negative-residual branch weakens contraction when $0 < w_{\text{eff}} < 1$; outward net deviation requires $w_{\text{eff}} < 0$, equivalent to $\eta > 1$ and $V_b < \tau^2(1 - 1/\eta)$.

Algorithm 1 One BDG-guided Euler step

Require: batch x_t , frozen v_θ and property map f , target y , $s > 0$, $\eta \geq 0$, $\tau > 0$, strength w , step h ,
start $t_{\min}$
1: $v \leftarrow v_\theta(x_t, t; M)$
2: **if** $t < t_{\min}$ **then**
3: **return** $\Pi(x_t + hv)$
4: **end if**
5: $m \leftarrow x_t + (1 - t)v_\theta(x_t, t; M)$ with gradients enabled
6: $F_i \leftarrow f(m_i)$; $\bar{F} \leftarrow B^{-1} \sum_i F_i$
7: $V_b \leftarrow (B - 1)^{-1} \sum_i (F_i - \bar{F})^2$ if $B > 1$, else τ^2
8: $e \leftarrow (V_b - \tau^2)/\tau^2$
9: $a_i \leftarrow \text{stopgrad}\{[(y - F_i) - \eta e(F_i - \bar{F})]/s^2\}$
10: $G \leftarrow \text{VJP}_{x_t}(F, a)$
11: $C_i \leftarrow w(1 - t)/\max(t, \varepsilon) G_i$
12: $C_i \leftarrow C_i \min(1, \|v_i\|/\max(\|C_i\|, \varepsilon))$
13: **return** $\Pi\{x_t + h(v + C)\}$

For molecules Π masks and centers; for sequences it clamps and renormalizes. At $w_{\text{eff}} \neq 0$, the scalar coefficient can be written $n_i = w_{\text{eff}}(y_{\text{eff}} - F_i)$ with $y_{\text{eff}} = (y + \eta e \bar{F})/w_{\text{eff}}$. At $w_{\text{eff}} = 0$ the original expression remains finite. This is a pointwise algebraic reduction to a batch-dependent gain and target; it does not prove equivalence to a fixed-strength, fixed-target plug-in sampler or predict the sign of a coverage difference.

Under the restrictive property-space dynamics $\dot{F}_i = (y - \bar{F}) - w_{\text{eff}}(F_i - \bar{F})$, variance obeys $\dot{V}_b = -2w_{\text{eff}}V_b$. If $\eta > 1$, the positive stationary variance is $\tau^2(1 - 1/\eta)$, below the requested setpoint. This assumes unit homogeneous response, no generator drift, no clipping and no projection; it supplies no convergence guarantee for the implemented sampler.

A.9 Implementation and Evidence Map

Table S19 maps the completed sweeps, including our VP benchmark, to code and 995 result-cell hashes. Archived pilots are excluded. Sample-level files are required for joint-yield and batch-aware inference. The manifest also records separate GC/CpG baseline checks and decoded target errors for the six molecular headline comparisons.

Table S19: Reproduction map, with paths relative to the repository root.

Component	Source
Molecular BDG / integration	proj1/src/guidance.py, proj1/src/sampling.py
FM training	proj1/scripts/train_fm.py
VP training	proj1/scripts/train_diffusion.py
Molecular result cells	results/v3/, stages v3 and v3abl, n2000
Sequence result cells	results/m2/, stages m2, m2abl, m2wsweep, m2win
Molecular protocol	docs/protocol/FULL_RUN_V3_PROTOCOL.md
Ablation protocol	docs/protocol/ABLATION_V3_PROTOCOL.md
Sequence protocol	docs/protocol/MODALITY2_V3_PROTOCOL.md
Tables and vector figures	paper/tools/build_results.py
Cell hashes and contrasts	paper/results_manifest_v9.json
Algebra checks	proj1/tests/test_bdg.py